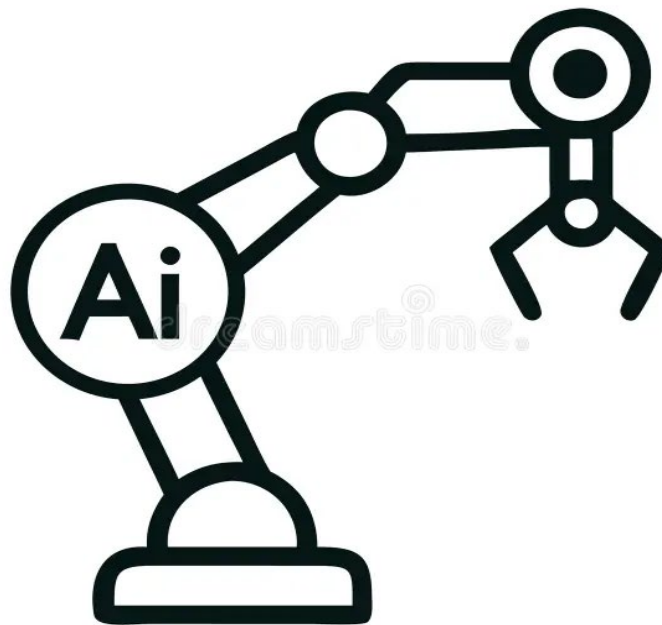


REPORT TITLE:
ROBOTIC ARM CONTROL SYSTEM
(PICK & PLACE ROBOTIC ARM WITH SDOF & HOMING FUNCTION)



MODULE: ROBOT SENSORS AND SIGNAL PROCESSING

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1. Project Specification

This report aims to design and build a (pick-and-place) robotic arm with a single degree of freedom (SDOF) and homing function. The system is built around a Raspberry Pi interfaced with various sensors and actuators. The angle of the robotic arm can also be controlled manually by a potentiometer. Below are the objectives of the system.

- A. Homing Functionality:** Press the homing button (Limit Switch) to reset the robotic arm base to a predefined “home” position. Once homed, the arm automatically moves to a start position with a 45° offset. A green LED signals that homing is complete.
- B. Manual Control:** The arm’s angle is adjustable by a potentiometer. The control range is mapped from a minimum of 45° to a maximum (up to 180°), with the current angle displayed on a monitor.
- C. Safety and Obstacle Detection:** An ultrasonic sensor continuously checks for obstacles. If no obstacle is detected within 15cm, a yellow LED will on to indicate safe for operation. Else a flashing red LED alerts the operator and disables the operation of the motors. Then operator must press the reset button and return it to home position.
- D. Task Execution:** Green LED to show system readiness. After the start button is pressed, the system picks up an object (simulated by moving the servomotor), then detects a placement position via an LDR sensor acting as an IR sensor. Gripper releases the object (simulated by moving the servomotor). After a successful placement, blue LED indicates task completion. One second after, the arm returns to its start position. (Note: System always check for obstacle detection and the normal operation keep repeating.)
- E. Emergency Handling:** An emergency stop button immediately halts all the operations at any times.

2. Hardware Design

2.1 Hardware Components

2.1.1 Controller

- RASPBERRY PI 3B+: Brain of the whole robotic arm system, handling GPIO and SPI communications.

2.1.2 Input Devices

- Ultrasonic Sensor (HC-SR04): For obstacle detection.
- LDR Sensor: Detects the placement position.
- Potentiometer: Control the angle of robotic arm manually.
- Push Button (SW1): As an emergency stop button.
- Push Button (SW2): To start the system.
- Push Button (SW3): To home position.
- Limit Switch (SW4): From home position to start position.

2.1.3 Output Devices

- Servo Motor (SG90): Moves the arm to desired angles (Gripper & Arm Base).
- Red LED: Alerts emergency condition or unsafe operation's surrounding (obstacle within 15 cm).
- Yellow LED: Represents the safe operational condition (obstacle without 15 cm).
- Blue LED: Confirms task completion.
- Green LED: Represents the system readiness or homing completion.

2.1.4 Other Components

- **MCP 3008 IC:** For SPI Communication, to convert analog sensor readings from the potentiometer and LDR sensor into digital data.
- **3D Printed Chassis:** For the servo motor to touch the limit switch automatically.
- **Breadboard**
- **Resistor**
- **Jumper Wire / Core Wire**

Name	Quantity	Component
SW1 SW2 SW3	3	Pushbutton
PR100K	1	100 kΩ Potentiometer
R4 R5 R6 R7	4	220 Ω Resistor
HC-SR04	1	Ultrasonic Distance Sensor (4-pin)
DRED	1	Red LED
DYELLOW	1	Yellow LED
DBLUE	1	Blue LED
DGREEN	1	Green LED
R2 = Voltage Divider	1	3.3 kΩ Resistor
R1	1	1.8 kΩ Resistor
R3	1	10 kΩ Resistor
MCP 3008	1	ADC IC
SERVOSG90	1	Positional Micro Servo
LDR	1	Light Dependent Sensor
Raspberry Pi 3b+	1	Raspberry Pi 3b+
SW4	1	Limit Switch

Fig 1: Circuit Component List

2.2 Pin Connections

2.2.1 GPIO Pin Allocation (BCM Numbering):

Inputs

- (SW1) Limit Switch (GPIO 17 with NO) Internal pull-up
- (SW2) Start Button (GPIO 22) Internal pull-up
- (SW3) Emergency Stop Button (GPIO 27) Internal pull-up
- (SW4) Homing Button (GPIO 16) Internal pull-up

Outputs:

- Ultrasonic Trigger (GPIO 20)
- Servo Motor (GPIO 18)
- Red LED (GPIO 5 with 220Ω resistor)
- Yellow LED (GPIO 6 with 220Ω resistor)
- Blue LED (GPIO 13 with 220Ω resistor)
- Green LED (GPIO 19 with 220Ω resistor)
- Ultrasonic Echo (GPIO 23 with Voltage Divider [R1=1.8KΩ & R2=3.3kΩ])

Voltage Calculation:

$$V_{out} = V_{in} \times (R_2 / (R_1 + R_2)), V_{in} = 5V$$

$$V_{out} = 5V \times (3300\Omega / (1800\Omega + 3300\Omega)) = 3.235V \approx 3.3V$$

Tip: If you don't have these exact resistor values, you can use close alternatives that provide a similar ratio (e.g. 1K Ω & 2K Ω).

SPI (MCP3008):

Power Connections:

- VDD and VREF → 3.3V
- AGND and DGND → GND

SPI Connections to Raspberry Pi:

- DIN (Data In) → MOSI (GPIO 10).
- DOUT (Data Out) → MISO (GPIO 9).
- CLK (Clock) → SCLK (GPIO 11).
- CS/SHDN (Chip Select) → CE0 (GPIO8).

Analog Sensor 1: Potentiometer

- Pin 1 → GND
- Pin 2 → MCP3008 CH0
- Pin 3 → 3.3V

Analog Sensor 2: Light Dependent Resistor (LDR)

- One side of LDR → 3.3V
- Another side of LDR → One side of 10kΩ resistor
- Another side of 10kΩ resistor → GND
- Between LDR and 10kΩ resistor → MCP3008 CH1.

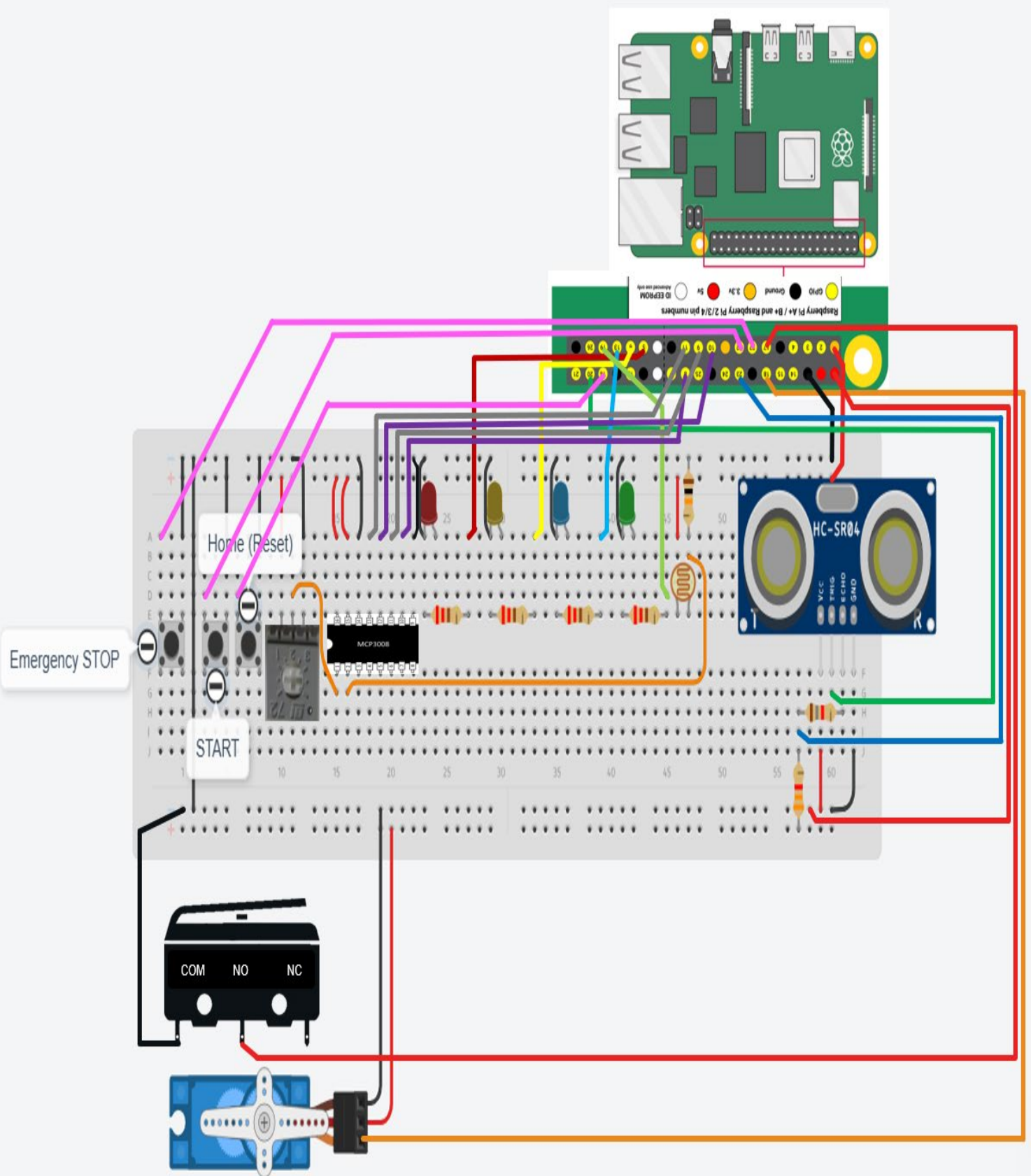


Fig 2: Wiring Diagram

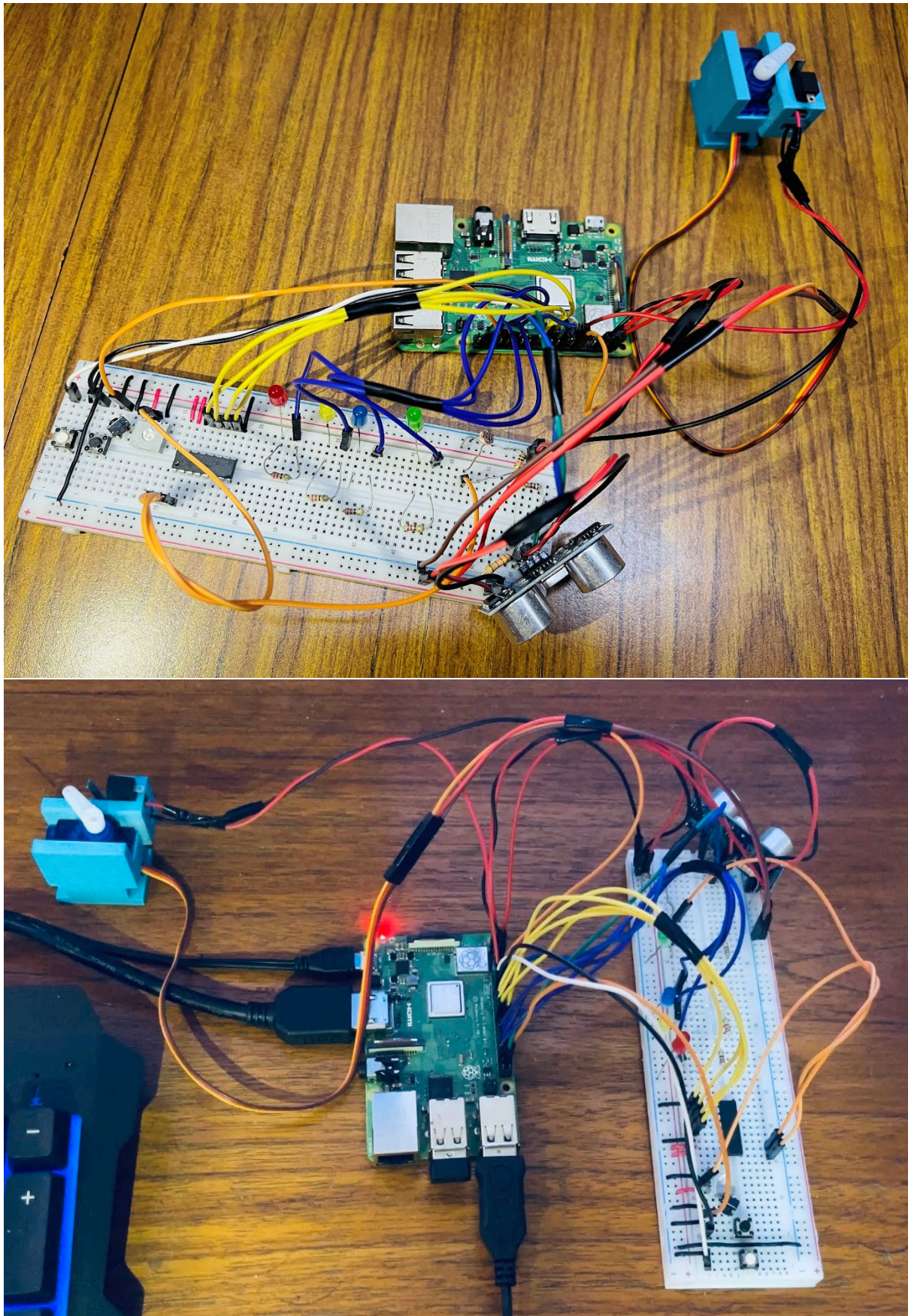


Fig 3: Completed Hardware Implementation

3. Software Design

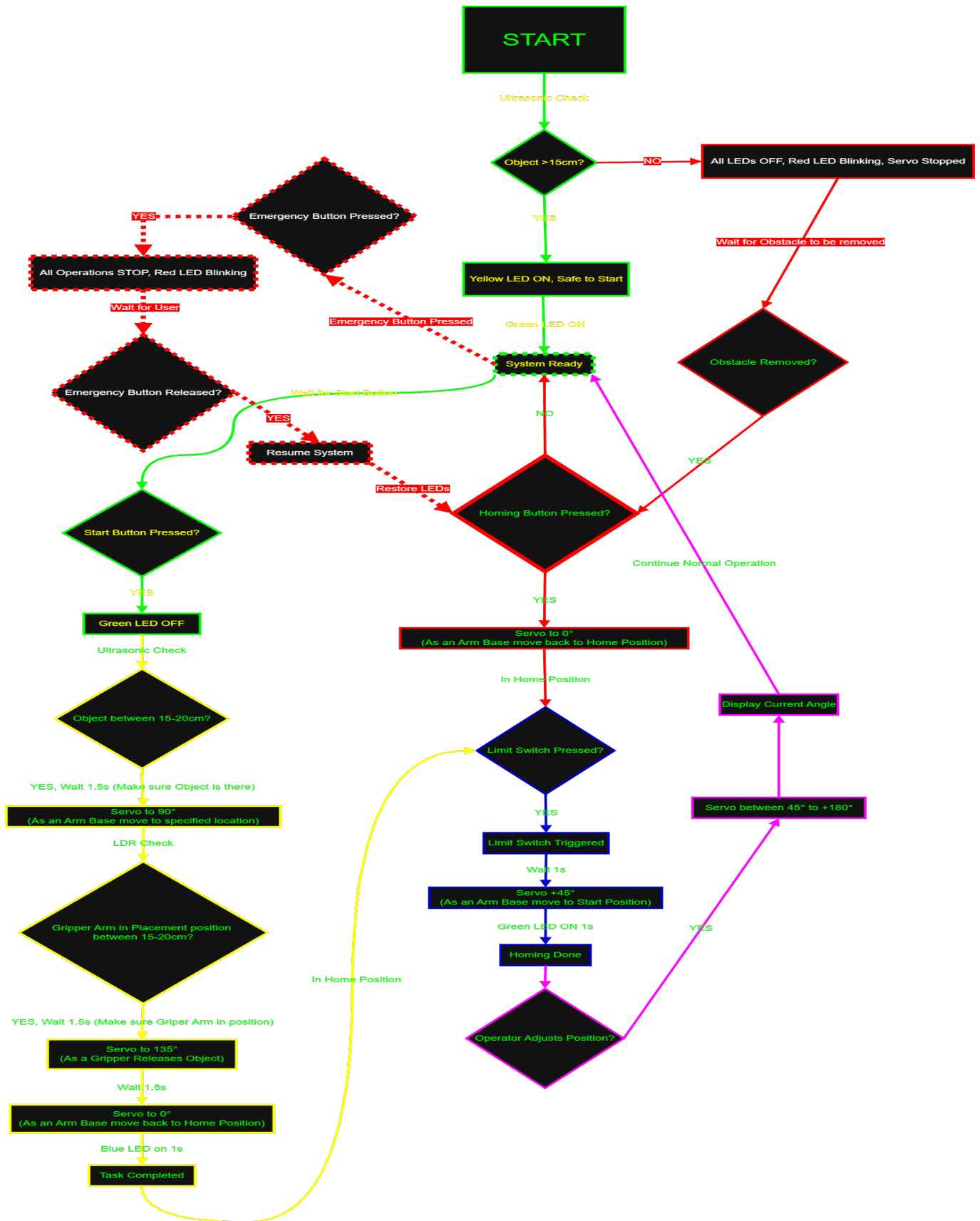


Fig 4: FLOWCHART

4. Implementation

4.1 Initialization - Before starting operation, the system the system must initialize all its hardware components.

- **GPIO setup:**

The Raspberry Pi's GPIO pins are configured using the RPi.GPIO library. All input pins (limit switch, start button, emergency stop button, homing button) are set with internal pull-up resistors. Output pins (ultrasonic trigger, servo, and LEDs) are set accordingly.

- **SPI Setup for Analog Sensors:**

The MCP3008 ADC is initialized using the spidev library to read analog values from the potentiometer and LDR sensor.

- **PWM Setup for Servo:**

The servo is controlled via PWM on a designated GPIO pin (GPIO 18). Note that we are using RPi.GPIO's PWM functionality rather than the pigpio library.

4.2 Startup Sequence and System Configuration - Once the hardware is initialized, the system enters a startup sequence.

- **System Ready Indication:**

The system turns on the Green LED to indicate readiness. The ultrasonic sensor is continuously checked to ensure that the environment is safe (e.g., no obstacle within 15 cm).

- **User Input:**

The system waits for the Start Button to be pressed. During this waiting period, the potentiometer continuously updates the servo's angle to allow manual adjustment (angles between 45° and 180°).

4.3 Object Detection & Handling

- **Object Pickup:**

The ultrasonic sensor detects whether an object is present at a distance between 15 and 20 cm. If detected, after a 1.5-second confirmation delay, the servo moves to 90° (simulating the gripper picking up the object).

- **Object Placement:**

Similarly, the LDR sensor is used to detect the object at the placement location. Upon confirmation, the servo moves to 135° to simulate releasing the object, then returns to 0° (with a Blue LED indicating task completion).

4.4 Homing Sequence - After the task completed, the system executes a homing sequence:

- **Homing Trigger:**

When the servo returns to 0°, it touches the limit switch. Then waits for 1 second before moving the servo an additional 45° to establish the starting (home) position.

- **Indicator:**

The Green LED is turned on during the homing procedure to confirm that homing is complete.

4.5 Safety Conditions and Emergency Handling

- **Continuous Safety Monitoring:**

The ultrasonic sensor is used throughout the operation to ensure no obstacles are within 15 cm. If an unsafe condition is detected, the system halts servo movement, and the Red LED blinks. Then must wait for user to press the Home button and reset to (home position --> start position).

- **Emergency Stop:**

At any time, if the Emergency Stop button is pressed, all operations immediately stop, and only the Red LED blinks until the emergency condition is cleared. Pressed again and resume the normal operation.

4.6 Potentiometer Control - The potentiometer allows manual adjustment of the servo's angle, and the system prints the current angle if the change exceeds 2° (due to the jitter issue). This ensures that the user receives real-time feedback on the arm's position.

4.7 Looping Cycle - The entire operation (startup, object handling, homing, safety checks, and emergency stop) is wrapped in a continuous loop, ensuring that the system is always monitoring and ready to operate in real-time.

5. Testing

5.1 Testing Procedures

In this section, describe how the system was tested. Testing was performed to validate the following aspects of the system.

5.1.1 Component Calibration:

- The ultrasonic sensor was calibrated using a test script to ensure reliable distance measurements.
- The LDR sensor and potentiometer were tested using the MCP3008 ADC to verify that their analog values were correctly mapped.
- The servo motor was verified for accurate movement at 0°, 45°, 90°, and 135°.

5.1.2 Functional Testing:

- Homing Function: Verified that when the servo reaches 0°, it touches the limit switch and triggers the homing routine.
- Object Pickup: Confirmed that the ultrasonic sensor correctly detects an object within the specified range, prompting the servo to move to 90°.
- Object Placement: Tested that the LDR sensor confirms the object at the placement location and the servo moves to 135°, then returns to 0° with the Blue LED indicating completion.
- Safety Monitoring: Ensured that if an obstacle is detected (≤ 15 cm), the system halts operations and displays a blinking Red LED.
- Emergency Stop: Checked that pressing the Emergency Stop button immediately halts all operations, and normal function resumes when the button is released.

5.2 Test Cases and Outcomes

Test Case	Expected Result	Actual Result
Homing Function	Once the home position is identified, the robotic arm base will move away from this position to establish a 45-degree offset between the home position and the start position, as illustrated in Figure 2. At the start position, a green LED will illuminate, indicating that the homing procedure is complete.	When servo reached 0° (Home Position), touching limit switch and moved to 45° (Start Position). Green LED on, monitor show homing completed.
Manual Control via Potentiometer	Users can use a potentiometer to control the position of the robotic arm base. It can control its position between 45° and +180°. The position is displayed on the monitor.	User able to control the angle via a potentiometer between 45° to +180°. Current angle displayed on the monitor.
Safety Monitoring	An ultrasonic sensor detects obstacles within 15 cm of the robotic arm. Obstacle detection: a) If no obstacle is detected, an LED will illuminate to indicate it is safe to start. b) If an obstacle is detected, a flashing LED alerts the operator and the power supply to the motors is disabled.	a) Same as expected result. b) Same as expected result too.
Return to Normal Operation	Once the issue is fixed, power is restored to the robotic arm motors. The operator then presses the home button.	After removed the obstacle, the flashing Red LED off, monitor inform operator to press home button. (Home button allow to be pressed at any time.)
Normal Operation	A green LED shows system readiness. Press 'start' button to begin. The system will receive an input when the object is ready to be picked up. The robotic arm, using a	Green LED on until user press the Start button then off. Ultrasonic detected the object (between 15 to 20cm), servo move to 90° as a gripper picked and moving the object.

	servomotor for the gripper, picks up and moves the object. An IR sensor circuit detects the placement position and releases the gripper. A blue or yellow LED lights up when the task is completed. After one second, the arm returns to the start position (note: the system should always be checking for obstacle detection) and repeats the operation. Incorporate an emergency 'stop' button to stop all operation.	An IDR sensor (as an IR sensor) detected the servo in the placement position, it moves to 135° as a gripper release and placing the object. Then servo back to 0°. A blue LED on, monitor show task is completed. In home position, it touches the limit switch. After 1 second, servo move to 45° (Start Position). System repeats the operation, and check for obstacle every time. Pressing the Emergency Stop button immediately halts all operations with only the Red LED blinking. Normal operation resumes after the button is released.
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5.3 Demo Description

- **Setup:**
The system was assembled on a breadboard with all wiring confirmed. The Raspberry Pi was connected to the sensors (ultrasonic, LDR, potentiometer) and output devices (servo and LEDs).
- **Operation:**
The system was powered on, and the Green LED indicated that it was ready. The operator pressed the Start Button, Green LED turn off and triggering the sequence of object pickup, placement, and homing (limit switch). Same time, the safety and emergency functions were also demonstrated by introducing an obstacle and activating the Emergency Stop button. Homing & emergency (push button) able to be pressed at any time.
- **Observations:**
The system behaved as expected across all test cases.

6. Discussion and Reflection:

6.1 Observations

- **System Reliability:**
The integration of multiple sensors (ultrasonic, LDR, potentiometer) and actuators (servo, LEDs) worked reliably to control the robotic arm's movements. The homing procedure and object handling operations were successfully executed, confirming that the hardware and software are well integrated. However, an issue was identified with using the potentiometer for manual control the inherent noise in the analog readings can cause the servo to jitter. To mitigate this, a filtering algorithm was implemented, which significantly reduced the jitter without compromising the system's responsiveness.

- **Response Time:**

The use of a 1.5-second delay for sensor confirmation improved the system's reliability, although it introduced slight delays in operation. In a production scenario, optimizing these delays could lead to faster response times. It also prevents any injuries from hitting the user when placing an object. Future optimizations could reduce these delays through advanced filtering or predictive algorithms, thereby enhancing both speed and safety.

- **Safety and Emergency:**

Continuous safety checks using the ultrasonic sensor effectively prevented the system from operating when obstacles were detected. The emergency stop function provided a critical safety layer by halting all operations immediately when triggered. Importantly, the system is designed to resume the last command instead of restarting completely once the emergency condition is cleared. This ensures that the robotic arm can quickly and safely resume its task without losing context, which is essential for maintaining operational continuity in a dynamic environment.

6.2 Improvements and Future Work

- **Sensor Fusion:**

In future iterations, integrating additional sensors (such as infrared arrays) could improve object detection accuracy and overall safety.

- **Response Optimization:**

Reducing the confirmation delay times could make the system more responsive without compromising reliability, possibly through advanced filtering or predictive algorithms.

- **Enhanced User Interface:**

Developing a graphical user interface (GUI) to display real-time sensor data and system status would enhance usability and facilitate troubleshooting.

- **Extended Functionality:**

Adding more degrees of freedom to the robotic arm and incorporating more complex motion planning algorithms would allow for a broader range of tasks and more dynamic operations.

- **Code Refinement:**

Further modularizing the code and implementing advanced error handling could improve system robustness and make future enhancements easier to integrate.

6.3 Reflections

Working on this project provided valuable insights into the challenges of integrating hardware and software in a real-time control system. The experience highlighted some few important point.

- **Robust Safety Mechanisms:**

Ensuring that the system can handle unexpected obstacles and emergencies is crucial for any robotic application. Safety protocols are essential because human life is the top priority, and the public places great importance on reliable safety measures.

- **Iterative Testing and Calibration:**

Fine-tuning sensor thresholds and delays was key to achieving reliable operation. For example, the issue of servo jitter when using the potentiometer

underscored the need for filtering algorithms. Iterative testing allowed for incremental adjustments that improved the overall stability of the system. Additionally, addressing button debouncing issues was critical for ensuring accurate and responsive user input. Careful design and logical arrangement of the software control sequences were also fundamental in attaining the desired performance.

- **System Integration:**

Successfully combining multiple sensors and actuators into a cohesive system required meticulous planning and thorough testing. For instance, using a voltage divider helped prevent damage to the GPIO pins, and integrating an MCP3008 chip for analog-to-digital conversion was crucial for obtaining accurate sensor readings.

In summary, there is a lot of considerations need to be considered when doing this kind of project. In order to meet the expected outcomes by thinking, planning, designing and testing the whole systems for robotic applications. But also provided a practical framework for addressing real-world challenges in system integration and control. The lessons learned here will be invaluable for me, if the future projects involving complex sensor networks and robotic control systems.

-----End – Of – Assignment-----